Korea Technology and Information Promotion Agency for SMEs

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| **Title of Proposed Technology** | Technology to calibrate traffic simulation by combining sparse real-world observations with urban-data-based parameter credibility models. | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| **Industrial Technology Classification** [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | *ICT* | | *Energy & Resources* | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | *Software* | *SW solution* | *New/Renewable Energy* | *Transportation system – Transportation infrastructure technology* |
| **Concept of Proposed Technology** | • **The proposed technology solves the problem of calibrating traffic simulations in cities with limited traffic measurements by combining sparse real-world observations with rich external urban and third-party data** (1).<br>• Technology provides methods and algorithms to build **credibility models** for key simulation parameters (OD matrices, driver behavior, signal control, etc.) based on (public available) urban and third-party data (e.g. land use, mobile data, socio-demographics, POIs, social media)<br>• Translates these credibility models into **prior density functions**, which express plausible parameter ranges per parameter and consistent parameter combinations, which can then be used to evaluate parameter fit. | | | |

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| | • Optimization methods and algorithms to calibrate the traffic simulation by **jointly optimizing**:<br>  ○ fit to sparse real-world observations (e.g., traffic counts, speeds, travel times, queues), and<br>  ○ consistency with the data-driven parameter credibility (priors).<br>• Optimization uses both **traffic simulation result estimation via AI** in the exploration step and complete simulation in the exploitation step to speed up the optimization process.<br>• Provides **robust, and realistic calibration** that prevents implausible parameter settings and enables more reliable traffic analysis, planning, and scenario testing even with sparse data. |
| **Rationale for Joint R&D Project** | • Practical gap: Current traffic simulation tools struggle to calibrate realistically when only sparse traffic measurements are available (2, 3); combining our calibration technology with an SME's operational solution is necessary to make it usable in real city operations.<br>• Research institute support: We contribute deep expertise in traffic simulation theory, uncertainty handling, and optimization (1, 4, 5)<br>• Korean SME collaboration: Established digital-twin and AI-based urban traffic operations platform, plus hands-on experience with traffic operators is necessary to bring it on the road.<br>• Joint R&D allows us to co-develop, embed, and test the new "parameter credibility–based" calibration methods directly inside an operational traffic analysis solution, ensuring that the technology is not only scientifically sound but also deployable.<br>• A Korean industrial partner is particularly valuable because Korea is a front-runner in AI-driven mobility and AX, with dense and complex urban networks that form ideal testbeds for advanced calibration methods and provide a strong basis for later commercialization in Asian mega-cities. |
| **Key Words** | 1. Traffic model calibration<br>2. Bayesian OD estimation<br>3. Parameter credibility models<br>4. Urban and third-party data fusion<br>5. AI-assisted optimization |

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level 3-4 | End of the Project | Level 6-7 |
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| R&D Objectives | • Develop a **data-driven methodology** to construct parameter credibility models (priors) for OD matrices, driver behavior, and signal control using urban and third-party data (e.g. land use, mobile data, socio-demographics, POIs, social media).<br>• Design and implement a **Bayesian-inspired calibration framework** that jointly optimizes fit to sparse real-world traffic observations and consistency with the parameter credibility models.<br>• Develop **AI-based surrogate models** to approximate traffic simulation outputs during the exploration phase, reducing the number of full simulation runs needed for calibration.<br>• Implement **optimization algorithms** that combine surrogate-based exploration with high-fidelity simulation exploitation for efficient calibration on large urban networks.<br>• Build and validate a **software prototype** (toolbox or module) that integrates these components and can be applied to real urban networks in collaboration with a Korean SME. | | | |
| R&D Target Specifications | • Achieve **traffic count and speed calibration accuracy** with a maximum normalized RMSE of **≤ 15%** on observed links, even when using **≤ 30%** of the sensor coverage normally required by state-of-the-art methods.<br>• Reduce the number of required full microscopic simulation runs during calibration by at least **50%** through AI-based surrogate models, while keeping calibration accuracy within **±5 percentage points** of a full-simulation-only baseline.<br>• Ensure that at least **80%** of calibrated key parameters (e.g. OD flows, car-following parameters, signal timings) lie within the **95% credible intervals** defined by the data-driven parameter credibility models. | | | |
| R&D Plan | Research Approach<br>• Center the research on the development and application of credibility models, which utilize prior information to inform and guide the calibration of traffic simulation parameters (1).<br>• Address the core technical challenge of calibrating traffic models under significant uncertainty caused by sparse and incomplete traffic count data (2, 3).<br>• Combine credibility models (1) with advanced technologies (e.g., machine learning, deep learning) (6, 7) to enable robust, data-driven decision-making in complex, real-world scenarios (3, 4).<br>Tasks and methods<br>• Elicit and formalize prior knowledge by engaging experts, reviewing literature, and analyzing historical data to establish the foundation for credibility models.<br>• Develop credibility models using statistical and Bayesian approaches to systematically incorporate prior information and quantify uncertainty in parameter calibration.<br>• Integrate credibility models into calibration workflows by embedding Bayesian inference directly into simulation calibration routines.<br>• Enhance calibration algorithms with optimization (such as metaheuristics), machine learning and deep learning techniques that leverage credibility information for efficient parameter search under uncertainty.<br>• Conduct sensitivity analysis to identify key parameters, using systematic variation and impact assessment to target where credibility models provide the most benefit.<br>• Validate and benchmark the approach through simulation experiments and cross-validation with synthetic and real-world data, comparing performance against traditional calibration methods, especially in sparse-data scenarios.<br>Within this plan we see key as milestones: Requirement & Knowledge Gathering, Credibility Model Developed, ML/DL integration into the optimization problem, Validation & Benchmarking. This is expected to be done within the timeline of 36 months, where in the first year we focus on the prior knowledge and credibility models development, the second year in the optimization problem with the integration of ML/DL and in the last year the refinement of the approach and its validation. | | | |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | • Design the overall calibration methodology, including mathematical formulation of parameter credibility models, Bayesian-inspired calibration framework, and AI-assisted optimization loop.<br>• Develop algorithms and prototype software for building credibility (prior) models from public and third-party urban data and for integrating these models into the calibration process.<br>• Design surrogate (AI-based) models for fast estimation of simulation results during the exploration phase, and optimization strategies that combine surrogate exploration with full simulation exploitation together with SME.<br>• Provide expertise and guidance on handling uncertainty and sparse data, define evaluation metrics and experimental setups, and lead the scientific analysis and documentation of results.<br>• Support integration of the developed modules into the SME's platform through APIs/specifications and contribute to joint dissemination (e.g. reports, publications, demnstrations). |
| **Expected R&R of Korean SME Partner** | • Provide real-world use cases, traffic simulation setups, and operational data from Korean cities, including sparse traffic measurements and available urban/third-party data relevant for calibration.<br>• Integrate the research institute's calibration and credibility-modeling components into their existing digital-twin and AI-based traffic operations platform, including software engineering and UI adaptation for operators.<br>• Run large-scale simulation experiments in the loop (using their simulation environment), support the development of AI-based result estimation, and contribute to performance tuning in realistic operational scenarios.<br>• Lead testing and validation with end users (e.g. local government traffic operators), collecting feedback on usability, reliability, and impact on decision-making.<br>• Prepare the path to commercialization by defining product requirements, packaging the joint results into a market-ready solution, and planning deployment and maintenance in Korean and international markets. |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | Technological Achievements:<br>• Develop innovative credibility models and enhanced calibration algorithms that integrate prior knowledge, enabling accurate and robust traffic simulation calibration even with sparse data.<br>• Create modular, interoperable software tools adaptable to various traffic simulation platforms.<br>Commercialization Potential:<br>• Deliver market-ready calibration technology, supporting licensing, service offerings, and broad adoption by traffic management agencies, urban planners, and simulation software providers.<br>• Provide a competitive advantage for SMEs and research partners through proprietary technology and specialized know-how.<br>Broader Industrial and Economic Impact:<br>• Improve traffic modeling and planning (e.g., traffic light plans), leading to more efficient urban mobility, better infrastructure investments, and reduced operational costs for cities and transport authorities.<br>• Strengthen collaboration between research institutions and industry, fostering innovation and capacity building in the smart mobility sector. |

## 6. References

Team:

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